

PHYS 357: Honours Quantum Physics 1

W.R.YCH.

August 31st, 2026

Abstract

Contents

1	Introduction	1
2	Mathematical Background	1
2.1	Taylor expansion	1
2.2	Basis	2
3	Stern-Gerlach Experiments	2
3.1	The classical picture	2
3.1.1	Angular momentum	2
3.1.2	Magnetic moment of a circulating charge	2
3.1.3	Gyromagnetic ratio	3
3.2	Spin	3
3.2.1	Intrinsic moment and the g-factor	3
3.2.2	Magnetons	4
3.2.3	Values for the electron, proton and neutron	4
3.3	The Stern-Gerlach apparatus	5
3.3.1	Force in an inhomogeneous field	5
3.3.2	Silver atoms	6
3.3.3	The classical prediction	6
3.3.4	The observed result	6
3.4	Sequential Stern-Gerlach experiments	7
3.4.1	Experiment 1: SGz then SGz	8
3.4.2	Experiment 2: SGz then SGx	8
3.4.3	Experiment 3: SGz, SGx, then SGz	9
3.4.4	Experiment 4: SGz, modified SGx, then SGz	9
4	Quantum States	9
4.1	State Space	9
4.1.1	Kets & Bras	9
4.1.2	Superposition	9
4.1.3	Completeness	10
4.2	Probability	10
4.2.1	Probability amplitudes	10
4.2.2	Normalization	11
4.2.3	Global phase	11
4.2.4	Expectation value	12
4.2.5	Uncertainty	12
4.3	Spin States	13
4.3.1	The state $ +\hat{x}\rangle$	13
4.3.2	Experiment 5: measuring S_y	13
4.3.3	The state $ +\hat{y}\rangle$ and the phase convention	13
4.4	Generalization to n -level systems	15
4.4.1	Orthonormality and completeness	15
4.4.2	Observables and the Born rule	15

4.4.3	Expectation value as an operator	16
5	Rotation of Basis States and Matrix Mechanics	16
5.1	States as column vectors	16
5.2	Operators as matrices	17
5.3	The spin operators	18
5.4	The rotation operator	19
5.4.1	Rotations about $\hat{z}$	20
5.4.2	Rotation as a change of basis	20
6	Change of Basis in n-Level Systems	21
6.1	Two bases for the same space	21
6.2	The transformation matrix	21
6.3	Eigenstates and eigenvalues	22
6.4	Operators in another basis	23
7	Generators of Rotations	23
7.1	Infinitesimal rotations	23
7.2	Functions of an operator	24
7.3	Hermitian operators	25
7.4	Rotations about any axis	26
8	Angular Momentum	27
8.1	Rotations about different axes	27
8.2	Compatible observables	28
8.3	Total angular momentum	28
A	Appendix	29
B	Solutions	29
C	Useful Links	32

1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Mathematical Background

2.1 Taylor expansion

Definition 1 (Taylor series). *A function f that is smooth near a equals the power series*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n, \quad (1)$$

where $f^{(n)}$ is the n th derivative. Expanding about $a = 0$ gives the Maclaurin series.

Result 1 (Series used in this course).

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad (2)$$

together with the binomial theorem $(1+u)^N = \sum_{n=0}^N \binom{N}{n} u^n$ for integer N .

Definition 2 (Geometric Series).

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (3)$$

Remark 1. Comparing the three series gives Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$, hence

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (4)$$

These two are used constantly when rotation matrices are written out.

Problem 1 (The exponential as a limit). Show that $\lim_{N \rightarrow \infty} (1 + x/N)^N = e^x$.

Remark 2. This limit is what lets a rotation be written as an exponential. A finite rotation is N infinitesimal ones stacked up,

$$R(\theta \hat{n}) = \lim_{N \rightarrow \infty} \left(\mathbb{1} - \frac{i}{\hbar} \frac{\theta}{N} \hat{n} \cdot \vec{S} \right)^N = e^{-i\theta \hat{n} \cdot \vec{S}/\hbar}. \quad (5)$$

2.2 Basis

Definition 3 (Basis). *A basis is a set of vectors that are linearly independent and span (generate) a vector space, acting as the fundamental building blocks from which every other vector in the space can be uniquely constructed.*

3 Stern-Gerlach Experiments

3.1 The classical picture

3.1.1 Angular momentum

Definition 4 (Orbital angular momentum). *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \quad (6)$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x - y plane, only the z component survives. On a circular path of radius r with speed v ,

$$L_z = mvr.$$

Definition 5 (Angular momentum of a rigid body). *For a rigid body¹ rotating about a fixed axis this is*

$$L = I_0\omega,$$

with I_0 the moment of inertia and ω the angular velocity.

Remark 3. *The moment of inertia I_0 measures how mass is distributed relative to the rotation axis.*

3.1.2 Magnetic moment of a circulating charge

Definition 6 (Magnetic moment). *A magnetic moment is a measure of the net circulation of electric charge. It is a vector quantity that determines the torque a magnet will experience in an external magnetic field.*

The magnetic moment of a point particle of mass m and charge q in a circular orbit of radius r and period T is

$$\mu = \frac{IA}{c} = \frac{1}{c} \left(\frac{qv}{2\pi r} \right) (\pi r^2) = \frac{q}{2mc} mvr = \frac{q}{2mc} L_z,$$

where I is the current and A is the area enclosed by the orbit. A single circulating charge counts as a steady current $I = q/T = qv/2\pi r$ because it passes any given point on the loop once per period.

¹A rigid body is one that does not deform under the action of external forces.

²Here c is in Gaussian units.

3.1.3 Gyromagnetic ratio

Definition 7 (Gyromagnetic ratio). *The gyromagnetic ratio γ is the constant of proportionality between the magnetic moment and the angular momentum,*

$$\vec{\mu} = \gamma \vec{L}. \quad (7)$$

For the circulating charge above,

$$\gamma = \frac{q}{2mc}. \quad (8)$$

Remark 4. *This is a purely classical result, since the charge is treated as a point particle on a definite orbit with a definite radius and speed, and nothing is quantized. It matters here because measuring $\vec{\mu}$ is how we measure $\vec{L}$, and it is what the Stern-Gerlach apparatus exploits.*

3.2 Spin

Definition 8 (Spin). *Spin $\vec{S}$ is an intrinsic angular momentum carried by a particle. It is a fixed property of the particle type, like charge or mass, and does not come from any motion through space. A stationary electron alone in empty space still has spin.*

Remark 5 (Where s comes from). *s is found by counting SG beams: $2s + 1$ beams. Silver gives 2 beams, so the electron has $s = \frac{1}{2}$. The deuteron (proton + neutron, spins aligned, $\frac{1}{2} + \frac{1}{2}$) gives 3 beams, so $s = 1$. Only $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ are allowed.*

Spin comes in addition to any orbital angular momentum $\vec{L}$ the particle has, not in place of it. By analogy with the classical result, it also carries a magnetic moment.

3.2.1 Intrinsic moment and the g-factor

The magnetic moment is still proportional to the angular momentum, but the gyromagnetic ratio picks up an extra dimensionless factor,

$$\vec{\mu} = \gamma \vec{S}, \quad \gamma = \frac{qg}{2mc}, \quad (9)$$

where g is the g-factor of the particle.

Definition 9 (g-factor). *The g-factor of a particle³ is the dimensionless number relating its magnetic moment to its spin angular momentum. It measures how much stronger or weaker the moment is than the classical value $\gamma = q/2mc$, which corresponds to $g = 1$.*

³Its value must be measured or derived from relativistic quantum theory.

3.2.2 Magnetons

Definition 10 (Planck constant). *The Planck constant h is the fundamental constant of quantum mechanics. It sets the scale at which quantization matters, and relates the energy of a photon to its frequency by $E = h\nu$. Its value is*

$$h \approx 6.626 \times 10^{-34} \text{ J s.}$$

It has units of $J \cdot s$, which is the same as angular momentum.

Definition 11 (Reduced Planck constant).

$$\hbar = \frac{h}{2\pi}. \quad (10)$$

It carries units of angular momentum, so $[\hbar] = [L_z] = \frac{kg \cdot m^2}{s}$. Its value is $\hbar = 1.055 \times 10^{-27} \text{ erg s} = 6.582 \times 10^{-16} \text{ eV s}$.

Writing $\vec{S} = \hbar(\vec{S}/\hbar)$ with $\vec{S}/\hbar$ dimensionless, the $\hbar$ joins the gyromagnetic ratio. The combination $q\hbar/2mc$ then carries the units and sets the natural size of the moment at a given mass scale.

Definition 12 (Bohr magneton). *The Bohr magneton is the natural unit of magnetic moment for an electron,*

$$\mu_B = \frac{|e|\hbar}{2m_e c}. \quad (11)$$

Definition 13 (Nuclear magneton). *The nuclear magneton is the same combination with the proton mass, and is the natural unit for nucleons,*

$$\mu_N = \frac{|e|\hbar}{2m_p c}. \quad (12)$$

Since $m_p/m_e \approx 1836$, we have $\mu_N \approx \mu_B/1836$.

Definition 14 (Nucleons). *The subatomic particles that make up the nucleus of an atom, specifically protons and neutrons.*

The electron moment is then written as

$$\vec{\mu}_e = g_e \mu_B \frac{\vec{S}}{\hbar}, \quad (13)$$

and similarly for the proton and neutron with μ_N in place of μ_B .

3.2.3 Values for the electron, proton and neutron

Particle	Charge q	Mass m	g -factor	Gyromagnetic ratio γ
Electron	$- e $	m_e	$g_e \approx 2$	$g_e \mu_B / \hbar$
Proton	$+ e $	m_p	$g_p \approx 5.59$	$g_p \mu_N / \hbar$
Neutron	0	m_p	$g_n \approx -3.83$	$g_n \mu_N / \hbar$

Table 1: Charges, masses, g -factors and gyromagnetic ratios.

Remark 6. The classical derivation gives $\mu = (q/2mc)L_z$, which vanishes when $q = 0$. The neutron is neutral yet has a nonzero moment, so its $\vec{\mu}$ cannot come from a single circulating point charge. This is a first sign that the classical picture fails.

Remark 7. Since $m_p \gg m_e$, we get $\mu_N \ll \mu_B$ and therefore $|\vec{\mu}_p| \ll |\vec{\mu}_e|$. Stern-Gerlach deflection of nuclei is much weaker than for electrons.

3.3 The Stern-Gerlach apparatus

Definition 15 (Stern-Gerlach apparatus). A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped to produce a strong field gradient along a chosen axis, here z . The gradient deflects each atom by an amount proportional to μ_z , and a screen records where it lands. Since $\vec{\mu} = \gamma\vec{S}$, the position on the screen is a direct measurement of S_z .

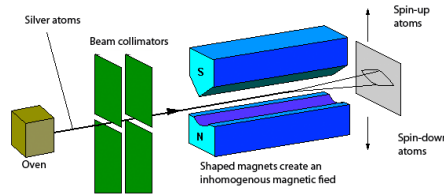


Figure 1: Illustration of the Stern-Gerlach apparatus.

3.3.1 Force in an inhomogeneous field

A magnetic moment in a field has potential energy

$$U = -\vec{\mu} \cdot \vec{B}, \quad (14)$$

so the force on it is

$$\vec{F} = -\vec{\nabla}U = \vec{\nabla}(\vec{\mu} \cdot \vec{B}), \quad F_z = -\frac{\partial U}{\partial z} = \mu_z \frac{\partial B_z}{\partial z}.$$

Definition 16 (Deflection). The displacement of an atom on the screen, perpendicular to the beam axis, produced by F_z acting over the length of the magnet. It is proportional to μ_z , so the pattern on the screen is a record of the moments in the beam.

Remark 8. A uniform field gives no force, only a torque. Thus, the apparatus needs a field gradient to deflect anything.

3.3.2 Silver atoms

Silver is used because its net orbital angular momentum vanishes, $\vec{L} = 0$, while $\vec{S} \neq 0$. The entire magnetic moment comes from a single unpaired valence electron, so the atom behaves as one electron spin. For the electron,

$$\mu_z = -\frac{g_e |e|}{2m_e c} S_z,$$

and therefore

$$F_z = \mu_z \frac{\partial B_z}{\partial z} = -g_e \frac{|e|}{2m_e c} S_z \frac{\partial B_z}{\partial z} = g_e \frac{|e|}{2m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z,$$

where the last equality assumes $\partial B_z / \partial z < 0$.

Result 2.

$$F_z \propto S_z.$$

The deflection of an atom measures the z component of its spin directly.

3.3.3 The classical prediction

Spin is a vector,

$$\vec{S} = (S_x, S_y, S_z)^T, \quad S_z = |\vec{S}| \cos \theta.$$

Remark 9. *The spins leaving the oven point in random directions, so θ takes every value in $[0, \pi]$ across the beam and $S_z = |\vec{S}| \cos \theta$ takes every value in $[-|\vec{S}|, |\vec{S}|]$. Since $F_z \propto S_z$, each atom is deflected differently and the screen should show a continuous smear, but it shows two spots.*

3.3.4 The observed result

Result 3 (Quantization of spin). *The electron spin angular momentum is quantized:*

$$S_z = \pm \frac{\hbar}{2}. \quad (15)$$

The beam splits into two spots.

Definition 17 (Spin quantum number). *The spin quantum number s labels the magnitude of a particle's spin and is fixed by the particle type. It takes the values $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$, and a measurement of S_z then gives one of $2s + 1$ outcomes,*

$$S_z = m\hbar, \quad m = -s, -s + 1, \dots, +s. \quad (16)$$

The electron has $s = \frac{1}{2}$, so $S_z = \pm \hbar/2$ and the beam splits in two. A spin-1 particle would split into three beams, $S_z = +\hbar, 0, -\hbar$.

Remark 10. *Everything in these notes is the $s = \frac{1}{2}$ case, so the state space is two-dimensional. Note also that S_z is a component, while the magnitude is $|\vec{S}| = \hbar\sqrt{s(s+1)}$, which for the electron is $\hbar\sqrt{3}/2$, not $\hbar/2$.*

Definition 18 (Basis state). *One of a set of states $\{|j\rangle\}$ that are mutually orthonormal and span the space, so that every state is a unique linear combination of them. For spin- $\frac{1}{2}$ the two outcomes of any one measurement, $\{|\pm\hat{n}\rangle\}$, form a basis.*

Definition 19 (Ket notation). *A ket $|\cdot\rangle$ labels a quantum state as a vector in a vector space. For electron spin the two measurement outcomes give the basis states*

$$|\uparrow\rangle \text{ for } S_z = +\frac{\hbar}{2}, \quad |\downarrow\rangle \text{ for } S_z = -\frac{\hbar}{2}.$$

The space is two-dimensional, so any spin state is a linear combination of the two.

Remark 11. *From here on we follow Townsend and write $|+\hat{z}\rangle \equiv |\uparrow\rangle$ and $|-\hat{z}\rangle \equiv |\downarrow\rangle$. The same labels work for any axis: $|\pm\hat{x}\rangle$ are the states with $S_x = \pm\hbar/2$.*

3.4 Sequential Stern-Gerlach experiments

Proposition 1. *We write SG_z for a device whose gradient points along z , and SG_x for the same device rotated by 90° about the beam axis.*

Definition 20 (Filter). *Each device sends atoms into two beams. A filter is a device that blocks one of the beams.*

Remark 12. *A beam straight from the oven always splits 50/50 in any SG device, whatever its orientation.*

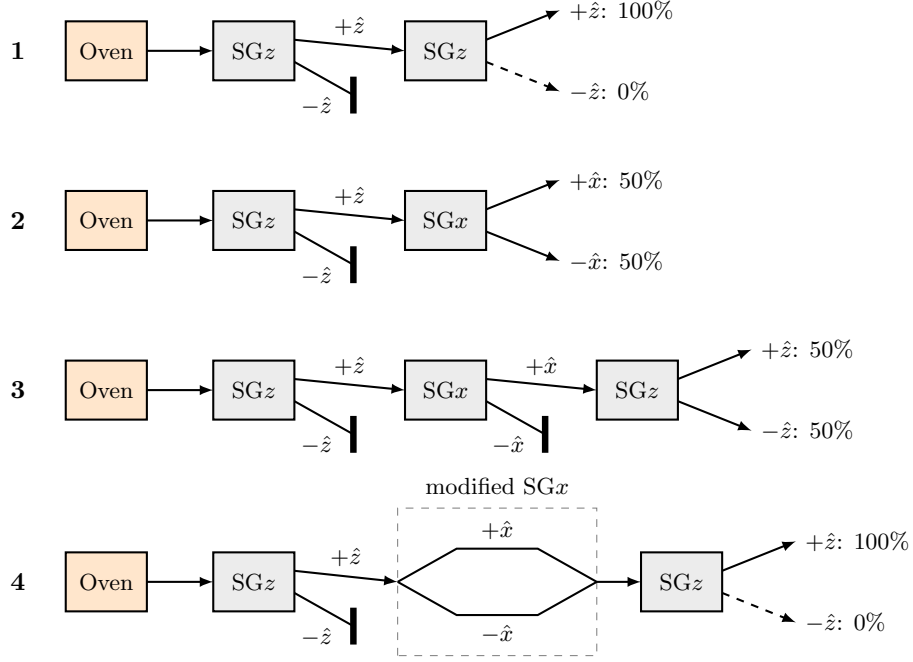


Figure 2: Sequential Stern-Gerlach experiments. Thick bars block a beam. Percentages are the fractions of atoms entering the last device. Every oven starts with N_0 atoms.

Remark 13. The lecture notes use N_0 differently, for the number of atoms entering the second device rather than leaving the oven. Their counts are therefore twice the ones below: for example Experiment 2 ends with $N_0/2$ in each S_x beam there, versus $N_0/4$ here.

3.4.1 Experiment 1: SG $_z$ then SG $_z$

Filter the $+\hat{z}$ beam and send it into a second SG $_z$. Every atom comes out $+\hat{z}$, none come out $-\hat{z}$. The measurement is repeatable: once an atom is found with $S_z = +\hbar/2$, measuring S_z again gives the same answer. The atom leaves the first device in the state $|+\hat{z}\rangle$. So we end with $N_0/2$ atoms in $|+\hat{z}\rangle$.

3.4.2 Experiment 2: SG $_z$ then SG $_x$

Filter the $+\hat{z}$ beam and send it into an SG $_x$. Half the atoms come out $+\hat{x}$ and half come out $-\hat{x}$. Knowing $S_z = +\hbar/2$ tells us nothing about S_x . We end up with $N_0/4$ in $|+\hat{x}\rangle$ and $N_0/4$ in $|-\hat{x}\rangle$.

3.4.3 Experiment 3: SG_z, SG_x, then SG_z

Filter $+\hat{z}$, then filter $+\hat{x}$, then measure S_z again. The beam splits 50/50 into $+\hat{z}$ and $-\hat{z}$ at the end, even though every atom was filtered to $+\hat{z}$ at the start. The S_x measurement wiped out the earlier S_z information. Blocking $+\hat{x}$ instead of $-\hat{x}$ gives the same 50/50 result. We end with $N_0/8$ in $|+\hat{z}\rangle$ and $N_0/8$ in $|-\hat{z}\rangle$.

Result 4. *An atom with a definite value of S_x does not have a definite value of S_z . Measuring S_x destroys what we knew about S_z , so no state has definite values of both at once.*

3.4.4 Experiment 4: SG_z, modified SG_x, then SG_z

Definition 21 (Modified SG device). *An SG device that splits the beam according to S_n and then recombines the two beams, so atoms with $S_n = \pm\hbar/2$ emerge at the same position even though they took two different paths.*

Filter $+\hat{z}$, send it through a modified SG_x with neither path blocked, then measure S_z . Every atom comes out $+\hat{z}$: 100%, not the 50/50 of Experiment 3. All $N_0/2$ atoms end in $|+\hat{z}\rangle$.

Result 5. *The modified SG_x cannot be sending $N_0/4$ through as $|+\hat{x}\rangle$ and $N_0/4$ as $|-\hat{x}\rangle$: Experiment 3 showed each of those alone gives 50/50 at the end. Instead the two paths interfere, as for water or light waves, cancelling the possibility of $-\hat{z}$. No path can be assigned to an atom.*

4 Quantum States

September 11-
18, 2026

4.1 State Space

4.1.1 Kets & Bras

Definition 22 (Bra). *Every ket (Def. 19) $|\psi\rangle$ has a matching bra $\langle\psi|$. Putting a bra in front of a ket gives a complex number, the inner product*

$$\langle\phi| |\psi\rangle \equiv \langle\phi|\psi\rangle \in \mathbb{C}.$$

Definition 23 (Orthonormal basis). *The basis kets are orthonormal:*

$$\langle+\hat{z}|+\hat{z}\rangle = \langle-\hat{z}|-\hat{z}\rangle = 1, \quad \langle+\hat{z}|-\hat{z}\rangle = \langle-\hat{z}|+\hat{z}\rangle = 0. \quad (17)$$

4.1.2 Superposition

Definition 24 (Superposition). *A general spin state is a linear combination of the basis kets,*

$$|\psi\rangle = c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle, \quad c_{\pm} \in \mathbb{C}. \quad (18)$$

The pair $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$ is a basis, and the coefficients $c_{\pm}$ are complex numbers.

4.1.3 Completeness

Definition 25 (Projector). *The operator $P_j = |j\rangle\langle j|$ built from a basis state. Acting on $|\psi\rangle = \sum_k c_k |k\rangle$ it keeps only the $|j\rangle$ part,*

$$P_j |\psi\rangle = c_j |j\rangle,$$

since $\langle j|k\rangle = \delta_{jk}$.

Proposition 2 (Completeness relation). *Since any state can be expanded in the basis $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$, the sum of the projectors onto the basis states is the identity:*

$$|+\hat{z}\rangle\langle+\hat{z}| + |-\hat{z}\rangle\langle-\hat{z}| = \mathbf{1}. \quad (19)$$

Inserting the identity in front of any state gives its expansion directly:

$$|\psi\rangle = (|+\hat{z}\rangle\langle+\hat{z}| + |-\hat{z}\rangle\langle-\hat{z}|) |\psi\rangle = \langle+\hat{z}|\psi\rangle |+\hat{z}\rangle + \langle-\hat{z}|\psi\rangle |-\hat{z}\rangle,$$

so the coefficients $c_{\pm}$ of the superposition are the inner products $\langle\pm\hat{z}|\psi\rangle$.

4.2 Probability

4.2.1 Probability amplitudes

Proposition 3. *To pull out c_+ , act on $|\psi\rangle$ with $\langle+\hat{z}|$:*

$$\langle+\hat{z}|\psi\rangle = c_+ \underbrace{\langle+\hat{z}|+\hat{z}\rangle}_{=1} + c_- \underbrace{\langle+\hat{z}|-\hat{z}\rangle}_{=0} = c_+.$$

The same steps with $\langle-\hat{z}|$ give c_- . So

$$c_{\pm} = \langle\pm\hat{z}|\psi\rangle. \quad (20)$$

Definition 26 (Probability amplitude). *The coefficient $c_{\pm} = \langle\pm\hat{z}|\psi\rangle$ is the probability amplitude for measuring $S_z = \pm\hbar/2$. The probability itself is the modulus squared,*

$$P(S_z = \pm\hbar/2) = |c_{\pm}|^2 = |\langle\pm\hat{z}|\psi\rangle|^2. \quad (21)$$

Principle 1 (Born rule). *The probability that a system in the state $|\psi\rangle$ is found in the state $|\phi\rangle$ is*

$$P(\phi|\psi) = |\langle\phi|\psi\rangle|^2 \in [0, 1]. \quad (22)$$

Remark 14. *Why is the Born rule true? There is no deeper derivation: it is a postulate that agrees with every experiment. As David Mermin put it, "shut up and calculate."*

4.2.2 Normalization

Proposition 4. *A system in $|\psi\rangle$ is found in $|\psi\rangle$ with certainty, so*

$$P(\psi|\psi) = |\langle\psi|\psi\rangle|^2 = 1.$$

To find what a bra equals to, write it with unknown coefficients, $\langle\psi| = c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|$. Then

$$\langle\psi|\psi\rangle = (c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|) (c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle) = c'_+ c_+ + c'_- c_-,$$

since the cross terms vanish by orthonormality. The two outcomes $S_z = \pm\hbar/2$ must have probabilities $|c_+|^2$ and $|c_-|^2$ that add to 1. For $\langle\psi|\psi\rangle$ to be that sum, each product $c'_\pm c_\pm$ must be real and equal to $|c_\pm|^2$. This forces $c'_\pm = c_\pm^*$.

Result 6. *Going from a ket to its bra, conjugate the coefficients:*

$$\langle\psi| = c_+^* \langle+\hat{z}| + c_-^* \langle-\hat{z}|. \quad (23)$$

In particular $\langle\psi|\pm\hat{z}\rangle = c_\pm^* = \langle\pm\hat{z}|\psi\rangle^*$. More generally, $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$.

Definition 27 (Normalization condition).

$$\langle\psi|\psi\rangle = \langle\psi|+\hat{z}\rangle \langle+\hat{z}|\psi\rangle + \langle\psi|-\hat{z}\rangle \langle-\hat{z}|\psi\rangle = |c_+|^2 + |c_-|^2 = 1. \quad (24)$$

In words, $P(S_z = +\hbar/2) + P(S_z = -\hbar/2) = 1$: the atom is always found in one of the two beams.

Example. Take

$$|\psi\rangle = \frac{1}{\sqrt{3}} |+\hat{z}\rangle + i\sqrt{\frac{2}{3}} |-\hat{z}\rangle.$$

The amplitude for $S_z = +\hbar/2$ is

$$\langle+\hat{z}|\psi\rangle = \frac{1}{\sqrt{3}} \underbrace{\langle+\hat{z}|+\hat{z}\rangle}_{=1} + i\sqrt{\frac{2}{3}} \underbrace{\langle+\hat{z}|-\hat{z}\rangle}_{=0} = \frac{1}{\sqrt{3}},$$

so $P(S_z = +\hbar/2) = 1/3$. In the same way $\langle-\hat{z}|\psi\rangle = i\sqrt{2/3}$, so $P(S_z = -\hbar/2) = |i|^2 \cdot 2/3 = 2/3$. The two add to 1, so the state is normalized.

4.2.3 Global phase

Proposition 5. *Take two kets that differ only by an overall phase, $|\psi\rangle = e^{i\alpha} |\psi'\rangle$ with $\alpha \in \mathbb{R}$. For any state $|\phi\rangle$,*

$$P(\phi|\psi) = |\langle\phi|\psi\rangle|^2 = |e^{i\alpha} \langle\phi|\psi'\rangle|^2 = |\langle\phi|\psi'\rangle|^2 = P(\phi|\psi'),$$

since $|e^{i\alpha}| = 1$.

Result 7. $|\psi\rangle$ and $e^{i\alpha} |\psi\rangle$ give the same probabilities for every possible measurement, so they describe the same physical state. An overall (global) phase has no physical meaning and can be dropped.

Remark 15. A relative phase is different: it sits between the two terms of a superposition, so it cannot be pulled out front and dropped. It has physical consequences, as the next subsection shows.

4.2.4 Expectation value

Definition 28 (Expectation value). *The expectation value of S_z in the state $|\psi\rangle$ is the average outcome weighted by the probabilities,*

$$\langle S_z \rangle = \left(\frac{\hbar}{2}\right) |\langle +\hat{z}|\psi \rangle|^2 + \left(-\frac{\hbar}{2}\right) |\langle -\hat{z}|\psi \rangle|^2. \quad (25)$$

Remark 16. *For N measurements on identically prepared atoms, with N_+ giving $+\hbar/2$ and N_- giving $-\hbar/2$, the measured average is*

$$\langle S_z \rangle_N = \frac{\hbar}{2} \frac{N_+ - N_-}{N},$$

which tends to $\langle S_z \rangle$ as $N \rightarrow \infty$.

Remark 17 (Shot noise). *For a finite number N of measurements, the sample average deviates from the expectation value by*

$$|\langle S_z \rangle_N - \langle S_z \rangle| \sim \frac{\Delta S_z}{\sqrt{N}}.$$

This spread is intrinsic, not a flaw of the apparatus: it appears whenever a state with a definite value of one observable (e.g. S_x for $|\hat{x}\rangle$) is measured for an incompatible observable (e.g. S_z).

4.2.5 Uncertainty

Definition 29 (Uncertainty). *The uncertainty ΔS_z is the standard deviation of the measurement outcomes,*

$$(\Delta S_z)^2 = \langle (S_z - \langle S_z \rangle)^2 \rangle = \langle S_z^2 \rangle - \langle S_z \rangle^2, \quad (26)$$

where

$$\langle S_z^2 \rangle = \left(\frac{\hbar}{2}\right)^2 |\langle +\hat{z}|\psi \rangle|^2 + \left(-\frac{\hbar}{2}\right)^2 |\langle -\hat{z}|\psi \rangle|^2. \quad (27)$$

Example. *For $|\hat{x}\rangle$, both probabilities are $1/2$, so*

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{2} - \frac{\hbar}{2} \cdot \frac{1}{2} = 0, \quad \langle S_z^2 \rangle = \left(\frac{\hbar}{2}\right)^2 \cdot \frac{1}{2} + \left(-\frac{\hbar}{2}\right)^2 \cdot \frac{1}{2} = \frac{\hbar^2}{4},$$

and therefore $\Delta S_z = \hbar/2$. The average is zero but every single measurement gives $\pm\hbar/2$, so the spread is maximal.

Example. *For $|\psi\rangle = \sqrt{\frac{1}{3}}|\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle$, the probabilities are $1/3$ and $2/3$, so*

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{3} - \frac{\hbar}{2} \cdot \frac{2}{3} = -\frac{\hbar}{6}, \quad \langle S_z^2 \rangle = \frac{\hbar^2}{4} \left(\frac{1}{3} + \frac{2}{3} \right) = \frac{\hbar^2}{4}.$$

Then

$$(\Delta S_z)^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{36} = \frac{9\hbar^2 - \hbar^2}{36} = \frac{2}{9}\hbar^2, \quad \Delta S_z = \frac{\sqrt{2}}{3}\hbar.$$

Remark 18. *For spin- $\frac{1}{2}$, $\langle S_z^2 \rangle$ is always equal to $\frac{\hbar^2}{4}$, since both outcomes square to the same number and the probabilities sum to 1.*

4.3 Spin States

4.3.1 The state $|+\hat{x}\rangle$

In Experiment 3, atoms in $|+\hat{x}\rangle$ enter an SG z and split 50/50:

$$P(S_z = \pm \frac{\hbar}{2}) = |\langle \pm \hat{z} | +\hat{x} \rangle|^2 = \frac{1}{2}.$$

So both amplitudes have modulus $1/\sqrt{2}$, and each can carry its own phase:

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} e^{i\delta_+} |+\hat{z}\rangle + \frac{1}{\sqrt{2}} e^{i\delta_-} |-\hat{z}\rangle = \frac{e^{i\delta_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle), \quad \delta = \delta_- - \delta_+,$$

with $\delta_{\pm} \in \mathbb{R}$. Dropping the global phase $e^{i\delta_+}$ leaves

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle).$$

Remark 19. *The relative phase δ survives, and Experiment 3 alone does not fix it. It is pinned down by combining more measurements (for example along y) with a phase convention.*

4.3.2 Experiment 5: measuring S_y

Definition 30 (Cyclic permutation). *A cyclic permutation of the axes relabels them as $x \rightarrow y$, $y \rightarrow z$, $z \rightarrow x$ (or the reverse). It is a rotation of the coordinate system, so no physics changes under it.*

Let Experiment 5 be a cyclic permutation of Experiment 3, SG x becomes SG z and SG z becomes SG y . Now the process is as follows: filter $+\hat{z}$, then measure S_y .

Proposition 6. *By symmetry it must split 50/50 as well, so every cross-axis overlap has the same modulus:*

$$|\langle +\hat{x} | +\hat{z} \rangle|^2 = |\langle +\hat{y} | +\hat{z} \rangle|^2 = |\langle -\hat{x} | +\hat{y} \rangle|^2 = \frac{1}{2}.$$

The same argument as for $|+\hat{x}\rangle$ then gives

$$|+\hat{y}\rangle = \frac{e^{i\gamma_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\gamma} |-\hat{z}\rangle), \quad \gamma = \gamma_- - \gamma_+.$$

4.3.3 The state $|+\hat{y}\rangle$ and the phase convention

Proposition 7. *Dropping the global phases, $\gamma_+ = \delta_+ = 0$, we obtain equations for x and y in terms of z :*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm e^{i\delta} |-\hat{z}\rangle), \quad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm e^{i\gamma} |-\hat{z}\rangle).$$

So using orthonormality of the $\hat{z}$ basis,

$$\langle +\hat{x} | +\hat{y} \rangle = \frac{1}{2} \left(1 + e^{i(\gamma-\delta)} \right), \quad |\langle +\hat{x} | +\hat{y} \rangle|^2 = \frac{1}{2} (1 + \cos(\gamma - \delta)).$$

Requiring this to equal 1/2 forces $\cos(\gamma - \delta) = 0$, that is $\gamma - \delta = \pm\pi/2$. The remaining freedom is convention.

Result 8 (Phase convention). *Take $\delta = 0$ and $\gamma = +\pi/2$. Then*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm |-\hat{z}\rangle), \quad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm i|-\hat{z}\rangle). \quad (28)$$

The bras are then

$$\langle \pm\hat{x} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \pm \langle -\hat{z} |), \quad \langle \pm\hat{y} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \mp i \langle -\hat{z} |). \quad (29)$$

Remark 20. *Complex coefficients are required, not just allowed. With real coefficients the relative phases could only be 0 or π , so $\gamma - \delta = \pm\pi/2$ would be impossible.*

Example. *For $|\psi\rangle = \sqrt{\frac{1}{3}}|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle$, compute $\langle S_y \rangle$.*

The amplitudes are

$$\begin{aligned} \langle +\hat{y} | \psi \rangle &= \sqrt{\frac{1}{3}} \langle +\hat{y} | +\hat{z} \rangle + i\sqrt{\frac{2}{3}} \langle +\hat{y} | -\hat{z} \rangle \\ &= \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + i\sqrt{\frac{2}{3}}(-i) \right) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + \sqrt{\frac{2}{3}} \right), \\ \langle -\hat{y} | \psi \rangle &= \sqrt{\frac{1}{3}} \langle -\hat{y} | +\hat{z} \rangle + i\sqrt{\frac{2}{3}} \langle -\hat{y} | -\hat{z} \rangle \\ &= \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + i\sqrt{\frac{2}{3}}(i) \right) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} - \sqrt{\frac{2}{3}} \right). \end{aligned}$$

Computing the probabilities

$$\begin{aligned} P(S_y = +\frac{\hbar}{2}) &= |\langle +\hat{y} | \psi \rangle|^2 = \frac{1}{2} + \frac{\sqrt{2}}{3}, \\ P(S_y = -\frac{\hbar}{2}) &= |\langle -\hat{y} | \psi \rangle|^2 = \frac{1}{2} - \frac{\sqrt{2}}{3}, \end{aligned}$$

which add to 1 as they must. Then

$$\langle S_y \rangle = \frac{\hbar}{2} \left[\left(\frac{1}{2} + \frac{\sqrt{2}}{3} \right) - \left(\frac{1}{2} - \frac{\sqrt{2}}{3} \right) \right] = \frac{\sqrt{2}}{3} \hbar.$$

The factor i that did nothing for S_z matters here: it is a relative phase, and it is exactly what tilts the spin toward $+\hat{y}$.

4.4 Generalization to n -level systems

4.4.1 Orthonormality and completeness

Definition 31 (Number of levels). *The number of levels n is the number of possible outcomes of one measurement. For a particle of spin s ,*

$$n = 2s + 1. \quad (30)$$

The basis states are labelled $|j\rangle$, $j = 1, \dots, n$, one per outcome.

Remark 21. *The label j counts outcomes, not axes. First choose what to measure, say S_x . Then $j = 1, 2$ are its two outcomes, $|1\rangle = |+\hat{x}\rangle$ and $|2\rangle = |-\hat{x}\rangle$. Measuring S_z instead gives a different basis, $|1\rangle = |+\hat{z}\rangle$ and $|2\rangle = |-\hat{z}\rangle$, but still $n = 2$.*

Remark 22. *A ket can carry more than one label. The labels are just names for the state, usually the quantum numbers that pin it down. For spin we write $|s, m\rangle$, where s fixes the particle and m gives the outcome $S_z = m\hbar$. For the electron, $|+\hat{z}\rangle = |\frac{1}{2}, +\frac{1}{2}\rangle$ and $|-\hat{z}\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle$.*

Example. *The deuteron has $s = 1$, so $n = 2(1) + 1 = 3$. Measuring S_z gives three outcomes, $+\hbar$, 0 and $-\hbar$, with basis states*

$$|1\rangle = |1, +1\rangle, \quad |2\rangle = |1, 0\rangle, \quad |3\rangle = |1, -1\rangle,$$

so $j = 1, 2, 3$ and an SGz device splits the beam into three.

Definition 32 (Orthonormal and complete basis). *The basis is orthonormal if*

$$\langle i | j \rangle = \delta_{ij},$$

and complete if every state can be expanded in it,

$$|\psi\rangle = \sum_j c_j |j\rangle, \quad c_j = \langle j | \psi \rangle, \quad \langle \psi | = \sum_j c_j^* \langle j|. \quad (31)$$

Completeness is equivalent to

$$\mathbb{1} = \sum_{j=1}^n |j\rangle \langle j|. \quad (32)$$

Example. *For $n = 2$, $\mathbb{1} = |+\hat{n}\rangle \langle +\hat{n}| + |-\hat{n}\rangle \langle -\hat{n}|$ for any $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$, so*

$$|\psi\rangle = \langle +\hat{x} | \psi \rangle |+\hat{x}\rangle + \langle -\hat{x} | \psi \rangle |-\hat{x}\rangle = \langle +\hat{y} | \psi \rangle |+\hat{y}\rangle + \langle -\hat{y} | \psi \rangle |-\hat{y}\rangle.$$

4.4.2 Observables and the Born rule

Definition 33 (Observable). *An observable A has outcomes a_j with states $|a_j\rangle$ forming a complete orthonormal set. For spin- $\frac{1}{2}$, A could be S_x , S_y or S_z , with outcomes $\pm\hbar/2$. By the Born rule,*

$$P(A = a_j) = |\langle a_j | \psi \rangle|^2. \quad (33)$$

Proposition 8. *Normalization $\langle\psi|\psi\rangle = 1$ is the statement that the probabilities sum to 1.*

Proof.

$$\begin{aligned}\langle\psi|\psi\rangle &= \left(\sum_i c_i^* \langle a_i|\right) \left(\sum_j c_j |a_j\rangle\right) = \sum_{i,j} c_i^* c_j \delta_{ij} \\ &= \sum_j |c_j|^2 = \sum_j P(A = a_j) = 1.\end{aligned}$$

Equivalently, inserting the identity, $\sum_j \langle\psi|a_j\rangle \langle a_j|\psi\rangle = \langle\psi| \left(\sum_j |a_j\rangle \langle a_j|\right) |\psi\rangle = \langle\psi|\psi\rangle$. $\square$

4.4.3 Expectation value as an operator

Proposition 9.

$$\langle A \rangle = \sum_j a_j |\langle a_j|\psi\rangle|^2 = \sum_j \langle\psi|a_j\rangle a_j \langle a_j|\psi\rangle = \langle\psi| \hat{A} |\psi\rangle, \quad (34)$$

where

$$\hat{A} = \sum_j a_j |a_j\rangle \langle a_j|. \quad (35)$$

Definition 34 (Uncertainty of an observable).

$$(\Delta A)^2 = \langle (A - \langle A \rangle)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2, \quad \langle A^2 \rangle = \langle\psi| \hat{A}^2 |\psi\rangle = \sum_j a_j^2 |c_j|^2. \quad (36)$$

5 Rotation of Basis States and Matrix Mechanics

September 21-
23, 2026

5.1 States as column vectors

Definition 35 (Representation of a state). *In the S_z basis, a ket is represented by the column of its components and a bra by the conjugate row, the adjoint:*

$$|\psi\rangle \xrightarrow{S_z} \psi = \begin{pmatrix} c_+ \\ c_- \end{pmatrix}, \quad \langle\psi| \xrightarrow{S_z} \psi^\dagger = (c_+^* \quad c_-^*), \quad (37)$$

with $c_\pm = \langle \pm \hat{z} | \psi \rangle$. Then

$$\langle\psi|\psi\rangle = \psi^\dagger \psi = |c_+|^2 + |c_-|^2. \quad (38)$$

Remark 23. Write $|\psi\rangle \xrightarrow{S_z} \psi$, not $|\psi\rangle = \psi$. The ket is basis-independent; the column depends on the basis.

Example.

$$|\pm\hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \quad |\pm\hat{y}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}.$$

Problem 2 (Other bases). In the S_x basis, $|\psi\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x}|\psi\rangle \\ \langle -\hat{x}|\psi\rangle \end{pmatrix}$, and similarly for S_y . Write the column representing $|+\hat{z}\rangle$ in the S_z , S_x and S_y bases.

Proposition 10 (Inner product as a matrix product). If $|\phi\rangle \xrightarrow{S_z} \phi$ and $|\psi\rangle \xrightarrow{S_z} \psi$, then the inner product is a row times a column,

$$\langle\phi|\psi\rangle = \phi^\dagger \psi, \quad (39)$$

and the Born rule reads $P(\phi|\psi) = |\phi^\dagger \psi|^2$.

Remark 24. If the state is given in another basis, first replace each basis ket by its S_z column and add. For $|\psi\rangle = a|+\hat{y}\rangle + b|-\hat{y}\rangle$,

$$\psi \xrightarrow{S_z} \frac{a}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} + \frac{b}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a+b \\ i(a-b) \end{pmatrix},$$

and $\psi^\dagger \psi = 1$ is worth checking before going further.

Problem 3 (A probability by matrix mechanics). Using matrix mechanics, find $P(S_x = +\hbar/2)$ for the state $|\psi\rangle = \sqrt{\frac{1}{5}}|+\hat{z}\rangle + i\sqrt{\frac{4}{5}}|-\hat{z}\rangle$.

5.2 Operators as matrices

Definition 36 (Projection operator). $P_{\pm\hat{n}} = |\pm\hat{n}\rangle\langle\pm\hat{n}|$, so that $\mathbb{1} = P_{+\hat{n}} + P_{-\hat{n}}$ and $P_{+\hat{z}}|\psi\rangle = c_+|+\hat{z}\rangle$.

Definition 37 (Matrix of an operator). In a basis $\{|j\rangle\}$, an operator A is represented by the matrix

$$A_{ij} = \langle i|A|j\rangle. \quad (40)$$

Proposition 11 (Operator acting on a state). Acting with an operator on a ket becomes a matrix times a column vector,

$$A|\psi\rangle \rightarrow A\psi.$$

Proof. The k th entry of the column representing $A|\psi\rangle$ is its overlap with the k th basis state, $\langle k|A|\psi\rangle$. Insert $\mathbb{1} = \sum_j |j\rangle\langle j|$ between A and $|\psi\rangle$:

$$\langle k|A|\psi\rangle = \sum_j \underbrace{\langle k|A|j\rangle}_{A_{kj}} \underbrace{\langle j|\psi\rangle}_{c_j} = \sum_j A_{kj} c_j.$$

This is exactly row k of the matrix A multiplied by the column $\psi = (c_1, c_2, \dots)^T$, which is the rule for matrix-times-vector multiplication. $\square$

Example.

$$P_{+\hat{z}} \xrightarrow{S_z} \begin{pmatrix} \langle +\hat{z} | P_{+\hat{z}} | +\hat{z} \rangle & \langle +\hat{z} | P_{+\hat{z}} | -\hat{z} \rangle \\ \langle -\hat{z} | P_{+\hat{z}} | +\hat{z} \rangle & \langle -\hat{z} | P_{+\hat{z}} | -\hat{z} \rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

so

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} c_+ \\ c_- \end{pmatrix} = \begin{pmatrix} c_+ \\ 0 \end{pmatrix}.$$

Proposition 12 (Properties of projectors). *A projector satisfies $P_{\hat{n}}^2 = P_{\hat{n}}$, since $(|\hat{n}\rangle \langle \hat{n}|)(|\hat{n}\rangle \langle \hat{n}|) = |\hat{n}\rangle \langle \hat{n}| \hat{n} \langle \hat{n}| = |\hat{n}\rangle \langle \hat{n}|$. In the S_z basis,*

$$P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbb{1} = P_{+\hat{z}} + P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Definition 38 (Adjoint). *Writing $A = \mathbb{1} A \mathbb{1} = \sum_{i,j} A_{ij} |i\rangle \langle j|$, the adjoint is*

$$A^\dagger = \sum_{i,j} A_{ij}^* |j\rangle \langle i|, \quad (A^\dagger)_{ij} = A_{ji}^*. \quad (41)$$

Remark 25. *For real entries the adjoint is just the transpose.*

Proposition 13 (Bra of a transformed state). *If $|\psi'\rangle = A |\psi\rangle$, then $\langle \psi' | = \langle \psi | A^\dagger$.*

Proposition 14 (Adjoint of a product).

$$(AB)^\dagger = B^\dagger A^\dagger. \quad (42)$$

Remark 26. *The order reverses, and $A^\dagger B^\dagger$ is a different matrix in general.*

5.3 The spin operators

Definition 39 (Spin operator). *The spin operator $\hat{S}_n$ is defined so that $\langle S_n \rangle = \langle \psi | \hat{S}_n | \psi \rangle$. Each outcome $\pm \hbar/2$ multiplies the projector onto its state $|\pm \hat{n}\rangle$:*

$$\hat{S}_n = \frac{\hbar}{2} (|+\hat{n}\rangle \langle +\hat{n}| - |-\hat{n}\rangle \langle -\hat{n}|), \quad \hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}. \quad (43)$$

Example. *In the S_z basis, $i, j \in \{+\hat{z}, -\hat{z}\}$ and*

$$\langle i | \hat{S}_n | j \rangle = \frac{\hbar}{2} \left(\langle i | +\hat{n} \rangle \langle +\hat{n} | j \rangle - \langle i | -\hat{n} \rangle \langle -\hat{n} | j \rangle \right), \quad \langle \pm \hat{n} | j \rangle = \langle j | \pm \hat{n} \rangle^*,$$

$$\hat{S}_n \xrightarrow{S_z} \begin{pmatrix} \langle +\hat{z} | \hat{S}_n | +\hat{z} \rangle & \langle +\hat{z} | \hat{S}_n | -\hat{z} \rangle \\ \langle -\hat{z} | \hat{S}_n | +\hat{z} \rangle & \langle -\hat{z} | \hat{S}_n | -\hat{z} \rangle \end{pmatrix}.$$

Problem 4 (The matrix of $\hat{S}_z$). *Find the matrix representation of $\hat{S}_z$ in the S_z basis.*

Problem 5 (The matrix of $\hat{S}_x$). Find the matrix representation of $\hat{S}_x$ in the S_z basis.

Problem 6 (The matrix of $\hat{S}_y$). Find the matrix representation of $\hat{S}_y$ in the S_z basis.

Definition 40 (Pauli matrices).

$$\hat{S}_j = \frac{\hbar}{2} \sigma_j, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (44)$$

all in the S_z basis. Each is its own adjoint, and each squares to $\mathbb{1}$, so $\hat{S}_j^2 = \frac{\hbar^2}{4} \mathbb{1}$ for every j .

Remark 27 (Shortcut). Instead of four separate brackets, write each projector as an outer product of a column with its conjugate row, for example

$$|+\hat{x}\rangle \langle +\hat{x}| \xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad |-\hat{x}\rangle \langle -\hat{x}| \xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

then subtract and multiply by $\hbar/2$.

Problem 7 (Expectation value and uncertainty by matrix mechanics). For $|\psi\rangle = \sqrt{\frac{1}{5}} |+\hat{y}\rangle + i\sqrt{\frac{4}{5}} |-\hat{y}\rangle$, find $\langle S_z \rangle$ and ΔS_z using matrix mechanics.

5.4 The rotation operator

Definition 41 (Rotation operator). $R(\theta\hat{n})$ rotates a state by θ about $\hat{n}$: $|\psi'\rangle = R(\theta\hat{n})|\psi\rangle$.

Example. $R(\frac{\pi}{2}\hat{y})$ takes $|\pm\hat{z}\rangle$ to $|\pm\hat{x}\rangle$ (up to phase), so

$$R\left(\frac{\pi}{2}\hat{y}\right)(c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle) = c_+|+\hat{x}\rangle + c_-|-\hat{x}\rangle.$$

Remark 28. $R(\theta\hat{n})$ is a linear operator: $R(c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle) = c_+R|+\hat{z}\rangle + c_-R|-\hat{z}\rangle$, which is what the equation above uses.

Proposition 15 (The bra of a rotated state). The bra of $|\psi'\rangle = R|\psi\rangle$ is $\langle\psi'| = \langle\psi|R^\dagger$.

Proof. Let $|+\hat{x}\rangle = R|+\hat{z}\rangle$ and $\langle+\hat{x}| = \langle+\hat{z}|R$. Two rotations by $\frac{\pi}{2}$ about $\hat{y}$ make one rotation by π , which takes $|+\hat{z}\rangle$ to $|-\hat{z}\rangle$, so

$$\langle+\hat{x}|+\hat{x}\rangle = \langle+\hat{z}|R\left(\frac{\pi}{2}\hat{y}\right)R\left(\frac{\pi}{2}\hat{y}\right)|+\hat{z}\rangle = \langle+\hat{z}|R(\pi\hat{y})|+\hat{z}\rangle \propto \langle+\hat{z}|-\hat{z}\rangle = 0.$$

This contradicts $\langle+\hat{x}|+\hat{x}\rangle = 1$. □

Definition 42 (Unitary operator). An operator U is unitary if $U^\dagger U = U U^\dagger = \mathbb{1}$.

Theorem 1. *Rotation operators are unitary:*

$$R^\dagger R = R R^\dagger = \mathbb{1}. \quad (45)$$

Proof. A rotation must preserve normalization: $\langle \psi' | \psi' \rangle = \langle \psi | R^\dagger R | \psi \rangle = 1$ for every $|\psi\rangle$ and every $R(\theta \hat{n})$, which forces $R^\dagger R = \mathbb{1}$. $\square$

Remark 29. *Write $U|\psi\rangle = c|\phi\rangle$ with $|\phi\rangle$ normalized. Then*

$$1 = \langle \psi | U^\dagger U | \psi \rangle = |c|^2 \langle \phi | \phi \rangle = |c|^2,$$

so $c = e^{i\varphi}$ is only a phase: a unitary maps a physical state to another physical state. In particular, the eigenvalues of a unitary satisfy $|c| = 1$, and U leaves the physical state unchanged only when $|\psi\rangle$ is an eigenstate.

5.4.1 Rotations about $\hat{z}$

Result 9. *A rotation about $\hat{z}$ cannot change S_z , so it can only multiply the S_z basis states by a phase:*

$$R(\phi \hat{z}) |\pm \hat{z}\rangle = e^{\mp i\phi/2} |\pm \hat{z}\rangle,$$

using $R = e^{-i\phi \hat{S}_z/\hbar}$ and Proposition 20.

Remark 30. *A full turn, $\phi = 2\pi$, gives $R = -\mathbb{1}$: a spin- $\frac{1}{2}$ state needs 4π to come back to itself. The sign is a global phase, so it is unobservable on its own.*

Remark 31. *In the S_z basis,*

$$R(\phi \hat{z}) \xrightarrow{S_z} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix},$$

but in other bases it is not diagonal.

Problem 8 ($R(\phi \hat{z})$ in the S_x basis). *Find the matrix representation of $R(\phi \hat{z})$ in the S_x basis.*

5.4.2 Rotation as a change of basis

Proposition 16. *Rotating the S_z basis by 90° about $\hat{y}$ gives the S_x basis, $R(\frac{\pi}{2} \hat{y}) |\pm \hat{z}\rangle = |\pm \hat{x}\rangle$.*

Result 10.

$$R(\frac{\pi}{2} \hat{y}) \xrightarrow{S_z} \begin{pmatrix} \langle +\hat{z} | +\hat{x} \rangle & \langle +\hat{z} | -\hat{x} \rangle \\ \langle -\hat{z} | +\hat{x} \rangle & \langle -\hat{z} | -\hat{x} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Remark 32. *The columns are the target kets $|\pm \hat{x}\rangle$ written as S_z columns. Likewise, the rotation taking $|\pm \hat{z}\rangle$ to $|\pm \hat{y}\rangle$ has columns $\frac{1}{\sqrt{2}}(1, \pm i)^T$.*

For a state given in the S_x basis, insert $\mathbb{1} = \sum_{\pm} |\pm\hat{x}\rangle \langle \pm\hat{x}|$ into each S_z amplitude:

$$\langle \pm\hat{z}|\psi\rangle = \langle \pm\hat{z}| + \hat{x}\rangle \langle +\hat{x}|\psi\rangle + \langle \pm\hat{z}| - \hat{x}\rangle \langle -\hat{x}|\psi\rangle,$$

which is the matrix above times the S_x column.

Result 11 (Change of basis).

$$\psi^z = R\left(\frac{\pi}{2}\hat{y}\right)\psi^x, \quad \psi^x = R^\dagger\left(\frac{\pi}{2}\hat{y}\right)\psi^z. \quad (46)$$

Problem 9 (Expectation value from the S_x basis). For $|\psi\rangle = \frac{1}{\sqrt{3}}|+\hat{x}\rangle + i\sqrt{\frac{2}{3}}|-\hat{x}\rangle$, find $\langle S_z \rangle = \psi^{z\dagger} \hat{S}_z \psi^z$ using matrix mechanics.

6 Change of Basis in n -Level Systems

6.1 Two bases for the same space

Take an n -level system with two orthonormal bases,

$$\begin{aligned} \{|j\rangle : j = 1, \dots, n\}, \quad & \{|\alpha\rangle : \alpha = 1, \dots, n\}, \\ \langle i|j\rangle = \delta_{ij}, \quad & \langle \alpha|\beta\rangle = \delta_{\alpha\beta}. \end{aligned}$$

Remark 33. The cross overlaps are not Kronecker deltas: $\langle j|\alpha\rangle \neq \delta_{j\alpha}$ in general, just as $\langle +\hat{x}| + \hat{z}\rangle = \frac{1}{\sqrt{2}}$ is neither 0 nor 1.

Both bases are complete, so the identity can be inserted either way:

$$|\psi\rangle = \mathbb{1}|\psi\rangle = \sum_{j=1}^n |j\rangle \langle j|\psi\rangle = \sum_{\alpha=1}^n |\alpha\rangle \langle \alpha|\psi\rangle. \quad (47)$$

The same ket then has two column representations,

$$|\psi\rangle \xrightarrow{|j\rangle} \psi^j = \begin{pmatrix} \langle 1|\psi\rangle \\ \langle 2|\psi\rangle \\ \vdots \\ \langle n|\psi\rangle \end{pmatrix}, \quad |\psi\rangle \xrightarrow{|\alpha\rangle} \psi^\alpha = \begin{pmatrix} \langle \alpha=1|\psi\rangle \\ \langle \alpha=2|\psi\rangle \\ \vdots \\ \langle \alpha=n|\psi\rangle \end{pmatrix}. \quad (48)$$

6.2 The transformation matrix

Inserting $\mathbb{1} = \sum_{\alpha} |\alpha\rangle \langle \alpha|$ into each $|j\rangle$ amplitude,

$$\langle j|\psi\rangle = \sum_{\alpha=1}^n \langle j|\alpha\rangle \langle \alpha|\psi\rangle,$$

which is a matrix times a column.

Result 12 (Change of basis).

$$\psi^j = U\psi^\alpha, \quad U_{j\alpha} = \langle j|\alpha\rangle. \quad (49)$$

The columns of U are the new basis kets $|\alpha\rangle$ written in the $|j\rangle$ basis.

Theorem 2. U is unitary: $U^\dagger U = UU^\dagger = \mathbb{1}$.

Proof. Using $(U^\dagger)_{\alpha j} = U_{j\alpha}^* = \langle \alpha|j\rangle$ and completeness of $\{|j\rangle\}$,

$$(U^\dagger U)_{\alpha\beta} = \sum_j \langle \alpha|j\rangle \langle j|\beta\rangle = \langle \alpha|\beta\rangle = \delta_{\alpha\beta}.$$

The same argument with completeness of $\{|\alpha\rangle\}$ gives $UU^\dagger = \mathbb{1}$. □

Remark 34. Normalization is basis-independent, as it must be:

$$\langle \psi|\psi\rangle = \psi^{j\dagger}\psi^j = \psi^{\alpha\dagger}U^\dagger U\psi^\alpha = \psi^{\alpha\dagger}\psi^\alpha = 1.$$

Example. With $|j\rangle = |\pm\hat{z}\rangle$ and $|\alpha\rangle = |\pm\hat{x}\rangle$, $U_{j\alpha} = \langle \pm\hat{z}|\pm\hat{x}\rangle$ is exactly $R(\frac{\pi}{2}\hat{y})$ from the previous section, and $\psi^j = U\psi^\alpha$ is $\psi^z = R\psi^x$.

6.3 Eigenstates and eigenvalues

Definition 43 (Eigenstate). A state $|a_i\rangle$ is an eigenstate of the operator $\hat{A}$ with eigenvalue a_i if

$$\hat{A}|a_i\rangle = a_i|a_i\rangle. \quad (50)$$

In matrix form this is the familiar $Mv = \lambda v$, with v the eigenvector.

Remark 35. A general operator maps a state to a different state, $\hat{A}|\psi\rangle = c|\phi\rangle$. Eigenstates are the special states that come back to themselves up to a number.

Example.

$$\hat{S}_z|\pm\hat{z}\rangle = \pm\frac{\hbar}{2}|\pm\hat{z}\rangle,$$

so $|\pm\hat{z}\rangle$ are the eigenstates of $\hat{S}_z$ and $\pm\hbar/2$ its eigenvalues, i.e. the measurement outcomes.

Proposition 17 (An operator is diagonal in its own eigenbasis).

$$A_{ij} = \langle a_i|\hat{A}|a_j\rangle = a_j\langle a_i|a_j\rangle = a_j\delta_{ij}, \quad (51)$$

so

$$\hat{A} \xrightarrow{|a_j\rangle} D = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{pmatrix}. \quad (52)$$

In particular $\langle a_i|\hat{A}|a_i\rangle = a_i$.

Example. For $\text{spin}-\frac{1}{2}$,

$$\hat{S}_z |+\rangle = \frac{\hbar}{2} |+\rangle, \quad \hat{S}_z |-\rangle = -\frac{\hbar}{2} |-\rangle.$$

Thus, in the S_z basis, $(S_z)_{ij} = \langle i | \hat{S}_z | j \rangle$ gives

$$\hat{S}_z \xrightarrow{S_z} \begin{pmatrix} \frac{\hbar}{2} & 0 \\ 0 & -\frac{\hbar}{2} \end{pmatrix} = \frac{\hbar}{2} \sigma_z.$$

6.4 Operators in another basis

Let $|j\rangle = |a_j\rangle$ be the eigenbasis of $\hat{A}$, so $\hat{A} \xrightarrow{|a_j\rangle} D$, and let $U_{j\alpha} = \langle a_j | \alpha \rangle$.

Proposition 18 (Similarity transformation).

$$\hat{A} \xrightarrow{|\alpha\rangle} A^\alpha = U^\dagger D U. \quad (53)$$

Proof. Insert the identity twice:

$$A_{\alpha\beta}^\alpha = \langle \alpha | \hat{A} | \beta \rangle = \sum_{i,j} \langle \alpha | a_i \rangle \langle a_i | \hat{A} | a_j \rangle \langle a_j | \beta \rangle = \sum_{i,j} (U^\dagger)_{\alpha i} D_{ij} U_{j\beta} = (U^\dagger D U)_{\alpha\beta}.$$

□

Result 13 (Expectation value in any basis).

$$\langle A \rangle = \langle \psi | \hat{A} | \psi \rangle = \psi^{j\dagger} D \psi^j = \psi^{\alpha\dagger} [U^\dagger D U] \psi^\alpha. \quad (54)$$

Either change the state to the eigenbasis ($\psi^j = U \psi^\alpha$) or change the operator to the state's basis ($A^\alpha = U^\dagger D U$). Both give the same number.

Problem 10 ($\hat{S}_z$ in the S_x basis). Using $U = R(\frac{\pi}{2}\hat{y})$, find the matrix of $\hat{S}_z$ in the S_x basis, and use it to redo the problem “Expectation value from the S_x basis” without changing the state's basis.

7 Generators of Rotations

September 28,
2026

7.1 Infinitesimal rotations

Proposition 19 (Composition of rotations about one axis). Rotations about the same axis add,

$$R(\theta_1 \hat{z}) R(\theta_2 \hat{z}) = R((\theta_1 + \theta_2) \hat{z}), \quad (55)$$

so a finite rotation can be split into $N \gg 1$ steps of size $\delta\theta = \theta/N$,

$$R(\theta \hat{z}) = [R(\frac{\theta}{N} \hat{z})]^N. \quad (56)$$

Taylor expanding a rotation about the identity,

$$R(x) = \mathbb{1} + \underbrace{\frac{dR}{dx}}_{\text{operator}} \Big|_{x=0} x + \mathcal{O}(x^2),$$

and writing the first-order operator as $-iJ_z/\hbar$, one small step is

$$R\left(\frac{\theta}{N}\hat{z}\right) = \mathbb{1} - \frac{i}{\hbar} J_z \frac{\theta}{N} + \mathcal{O}\left(\frac{1}{N^2}\right). \quad (57)$$

Result 14. *Using the limit from the problem “The exponential as a limit”,*

$$R(\theta\hat{z}) = \lim_{N \rightarrow \infty} \left(\mathbb{1} - \frac{i}{\hbar} J_z \frac{\theta}{N} + \mathcal{O}\left(\frac{1}{N^2}\right) \right)^N = e^{-iJ_z\theta/\hbar}. \quad (58)$$

Definition 44 (Generator of rotations). *The operator J_z is the generator of rotations about $\hat{z}$. It has units of $\hbar$, which are those of angular momentum,*

$$[J_z] = [\hbar] = [\vec{L}] = [\vec{r} \times \vec{p}] = [\vec{J}],$$

so $\vec{J}$ is identified as the angular momentum operator.

Remark 36. *For $\text{spin}=\frac{1}{2}$, $J_z = S_z$, so $R(\theta\hat{z}) = e^{-iS_z\theta/\hbar}$, in agreement with the rotations about $\hat{z}$ found earlier.*

7.2 Functions of an operator

Proposition 20 (Function of an operator on an eigenstate). *If $\hat{A}|a\rangle = a|a\rangle$ and f has a Taylor series, then*

$$f(\hat{A})|a\rangle = f(a)|a\rangle. \quad (59)$$

Proof. Applying $\hat{A}$ repeatedly gives $\hat{A}^n|a\rangle = a^n|a\rangle$. The operator $f(\hat{A})$ is defined by its Taylor series, so term by term

$$f(\hat{A})|a\rangle = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \hat{A}^n|a\rangle = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} a^n|a\rangle = f(a)|a\rangle.$$

□

Example. *With $\hat{A} = \hat{S}_z$, $\hat{S}_z|\pm\hat{z}\rangle = \pm\frac{\hbar}{2}|\pm\hat{z}\rangle$ and $f(x) = e^{-i\phi x/\hbar}$,*

$$\begin{aligned} R(\phi\hat{z})|\pm\hat{z}\rangle &= \left(\mathbb{1} - \frac{i\phi}{\hbar} \hat{S}_z + \frac{1}{2!} \left(\frac{-i\phi}{\hbar} \right)^2 \hat{S}_z^2 + \dots \right) |\pm\hat{z}\rangle \\ &= \left(1 \mp \frac{i\phi}{2} + \frac{1}{2!} \left(\mp \frac{i\phi}{2} \right)^2 + \dots \right) |\pm\hat{z}\rangle = e^{\mp i\phi/2} |\pm\hat{z}\rangle. \end{aligned}$$

So $|\pm\hat{z}\rangle$ are eigenstates of $R(\phi\hat{z})$ with eigenvalues $e^{\mp i\phi/2}$.

Proposition 21 (Spectral form). *Inserting $\mathbb{1} = \sum_j |a_j\rangle\langle a_j|$ on the right of $f(\hat{A})$ and using Proposition 20 on each term,*

$$f(\hat{A}) = f(\hat{A}) \sum_j |a_j\rangle\langle a_j| = \sum_j f(a_j) |a_j\rangle\langle a_j|. \quad (60)$$

With $f(x) = x$ this is $\hat{A} = \sum_j a_j |a_j\rangle\langle a_j|$, as found for the expectation value.

7.3 Hermitian operators

Definition 45 (Commutator).

$$[A, B] = AB - BA. \quad (61)$$

A and B commute if $[A, B] = 0$.

Definition 46 (Hermitian operator). *An operator A is Hermitian if $A = A^\dagger$.*

Proposition 22. *The generator J_z is Hermitian.*

Proof. Let $|\psi'\rangle = R(\theta\hat{z})|\psi\rangle$. Normalization must be preserved, so

$$\langle\psi'|\psi'\rangle = \langle\psi|R^\dagger(\theta\hat{z})R(\theta\hat{z})|\psi\rangle = \langle\psi|e^{iJ_z^\dagger\theta/\hbar}e^{-iJ_z\theta/\hbar}|\psi\rangle = 1.$$

To first order in θ ,

$$R^\dagger R = \mathbb{1} + \frac{i\theta}{\hbar} (J_z^\dagger - J_z) + \mathcal{O}(\theta^2) = \mathbb{1},$$

which forces $J_z^\dagger = J_z$. □

Remark 37. *The two exponentials may only be combined into one as $e^{i(J_z^\dagger - J_z)\theta/\hbar}$ if $[J_z, J_z^\dagger] = 0$. This holds once $J_z = J_z^\dagger$, since every operator commutes with itself.*

Principle 2. *Every observable A is represented by a Hermitian operator, $A = A^\dagger$.*

Proposition 23. *The eigenvalues of a Hermitian operator are real.*

Proof. Let $A|a\rangle = a|a\rangle$. Taking the adjoint gives $\langle a|A^\dagger = a^*\langle a|$. Then

$$a\langle a|a\rangle = \langle a|A|a\rangle = \langle a|A^\dagger|a\rangle = a^*\langle a|a\rangle,$$

so $a = a^*$. Measurement outcomes are therefore real, as they must be. □

7.4 Rotations about any axis

The same construction works for every axis: $R(\phi\hat{n}) = e^{-i\phi\hat{S}_n/\hbar}$, whose eigenstates are the eigenstates $|\pm\hat{n}\rangle$ of $\hat{S}_n$, with $\hat{S}_n|\pm\hat{n}\rangle = \pm\frac{\hbar}{2}|\pm\hat{n}\rangle$.

Proposition 24 (Spectral form of a rotation).

$$R(\phi\hat{n}) = e^{-i\phi/2} |+\hat{n}\rangle \langle +\hat{n}| + e^{i\phi/2} |-\hat{n}\rangle \langle -\hat{n}|, \quad \hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}. \quad (62)$$

Proof. Apply the spectral form to $f(x) = e^{-i\phi x/\hbar}$ with $\hat{A} = \hat{S}_n$ and eigenvalues $\pm\hbar/2$. $\square$

Remark 38. To get the matrix of $R(\phi\hat{n})$ in any basis $\{|i\rangle\}$, take matrix elements of the spectral form,

$$\langle i|R(\phi\hat{n})|j\rangle = \langle i|+\hat{n}\rangle e^{-i\phi/2} \langle +\hat{n}|j\rangle + \langle i|-\hat{n}\rangle e^{i\phi/2} \langle -\hat{n}|j\rangle.$$

This is the product $VDV^\dagger$, where $D = \text{diag}(e^{-i\phi/2}, e^{i\phi/2})$ and V has the eigenkets $|\pm\hat{n}\rangle$, written in the $\{|i\rangle\}$ basis, as its columns. It is the similarity transformation with the roles of the two bases swapped.

Proposition 25 ($R(\phi\hat{x})$ in the S_z basis).

$$R(\phi\hat{x}) \xrightarrow{S_z} \begin{pmatrix} \cos \frac{\phi}{2} & -i \sin \frac{\phi}{2} \\ -i \sin \frac{\phi}{2} & \cos \frac{\phi}{2} \end{pmatrix}. \quad (63)$$

Proof. The columns of V are $|\pm\hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}}(1, \pm 1)^T$, so $V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, which is real and symmetric, hence $V^\dagger = V$. Then

$$\begin{aligned} R(\phi\hat{x}) &\xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} e^{-i\phi/2} + e^{i\phi/2} & e^{-i\phi/2} - e^{i\phi/2} \\ e^{-i\phi/2} - e^{i\phi/2} & e^{-i\phi/2} + e^{i\phi/2} \end{pmatrix}. \end{aligned}$$

Euler's formula gives $e^{-i\phi/2} + e^{i\phi/2} = 2 \cos \frac{\phi}{2}$ and $e^{-i\phi/2} - e^{i\phi/2} = -2i \sin \frac{\phi}{2}$. $\square$

Remark 39. This is the same matrix as $R(\phi\hat{z})$ in the S_x basis.

Proposition 26 ($R(\phi\hat{y})$ in the S_z basis).

$$R(\phi\hat{y}) \xrightarrow{S_z} \begin{pmatrix} \cos \frac{\phi}{2} & -\sin \frac{\phi}{2} \\ \sin \frac{\phi}{2} & \cos \frac{\phi}{2} \end{pmatrix}. \quad (64)$$

Proof. Now the columns of V are $|\pm\hat{y}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}}(1, \pm i)^T$, so

$$V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \quad V^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix},$$

and this time $V^\dagger \neq V$. Then

$$\begin{aligned} R(\phi\hat{y}) &\xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} e^{-i\phi/2} + e^{i\phi/2} & -i(e^{-i\phi/2} - e^{i\phi/2}) \\ i(e^{-i\phi/2} - e^{i\phi/2}) & e^{-i\phi/2} + e^{i\phi/2} \end{pmatrix}. \end{aligned}$$

With $e^{-i\phi/2} - e^{i\phi/2} = -2i \sin \frac{\phi}{2}$, the off-diagonal entries become $-i(-2i)\frac{1}{2} \sin \frac{\phi}{2} = -\sin \frac{\phi}{2}$ and $i(-2i)\frac{1}{2} \sin \frac{\phi}{2} = +\sin \frac{\phi}{2}$. $\square$

Remark 40. This matrix is real, so $R(\phi\hat{y})$ never introduces factors of i into an S_z column. At $\phi = \pi$ it gives $R(\pi\hat{y})|+\hat{z}\rangle = |-\hat{z}\rangle$, the flip used in the proof that bras carry $R^\dagger$.

Remark 41 (Phase caveat). At $\phi = \pi/2$,

$$R\left(\frac{\pi}{2}\hat{y}\right) \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix},$$

whose second column is $-|\hat{x}\rangle$. The change-of-basis matrix $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ from the earlier section was defined by requiring $R|-\hat{z}\rangle = |\hat{x}\rangle$ exactly, so the two differ by a phase on one basis state. Both are valid, since each basis ket is only fixed up to a phase.

8 Angular Momentum

September 28,
2026

8.1 Rotations about different axes

Write $R_x = R(\theta\hat{x}) = e^{-i\theta J_x/\hbar}$ and $R_y = R(\theta\hat{y}) = e^{-i\theta J_y/\hbar}$.

Proposition 27. Rotations about different axes do not commute:

$$R_x R_y \neq R_y R_x.$$

Remark 42. Take a textbook and rotate it by 90° about $\hat{x}$ and then about $\hat{y}$, using the right-hand rule. Repeat from the same starting position with $\hat{y}$ first and then $\hat{x}$. The spine ends up pointing in different directions.

Result 15 (Angular momentum commutation relations).

$$[J_x, J_y] = i\hbar J_z, \quad [J_y, J_z] = i\hbar J_x, \quad [J_z, J_x] = i\hbar J_y, \quad (65)$$

or compactly

$$\vec{J} \times \vec{J} = i\hbar \vec{J}, \quad \vec{J} = J_x \hat{x} + J_y \hat{y} + J_z \hat{z}. \quad (66)$$

8.2 Compatible observables

Theorem 3. *If $[A, B] = 0$, then A and B have a common set of eigenstates.*

Proof. Let $A|a\rangle = a|a\rangle$. Since $AB = BA$,

$$A(B|a\rangle) = BA|a\rangle = a(B|a\rangle),$$

so $B|a\rangle$ is also an eigenstate of A with eigenvalue a .

Nondegenerate case. If $|a\rangle$ is the only state with eigenvalue a , then $B|a\rangle$ must be proportional to it, $B|a\rangle = b|a\rangle$, and $|a\rangle$ is an eigenstate of both.

Degenerate case. Suppose n states $|a, i\rangle$, $i = 1, \dots, n$, share the eigenvalue a , so $A|a, i\rangle = a|a, i\rangle$. By the argument above, B maps this subspace into itself, so in the $\{|a, i\rangle\}$ basis it is an $n \times n$ matrix,

$$B \xrightarrow{|a, i\rangle} \begin{pmatrix} B_{11} & \cdots & B_{1n} \\ \vdots & \ddots & \vdots \\ B_{n1} & \cdots & B_{nn} \end{pmatrix}, \quad B_{ij} = \langle a, i | B | a, j \rangle.$$

Diagonalizing this matrix gives eigenstates

$$|a, b_j\rangle = \sum_i |a, i\rangle \langle a, i | a, b_j \rangle,$$

which are linear combinations of the $|a, i\rangle$, and therefore satisfy

$$A|a, b_j\rangle = a|a, b_j\rangle, \quad B|a, b_j\rangle = b_j|a, b_j\rangle.$$

□

Definition 47 (Compatible observables). *Two observables are compatible if they commute, so that they can be simultaneously diagonalized. Otherwise they are incompatible.*

Remark 43. *Since $[J_x, J_y] = i\hbar J_z \neq 0$, the components of angular momentum are incompatible and have no common eigenbasis. This is the operator version of Experiment 3: no state has definite S_x and S_z at once.*

8.3 Total angular momentum

Definition 48 (Square of the angular momentum).

$$\vec{J}^2 = J_x^2 + J_y^2 + J_z^2. \quad (67)$$

Proposition 28. $\vec{J}^2$ commutes with every component of $\vec{J}$; in particular

$$[\vec{J}^2, J_z] = 0. \quad (68)$$

Proof. Compute $[J_z, \vec{J}^2]$ term by term with $[A, BC] = B[A, C] + [A, B]C$:

$$\begin{aligned}[J_z, J_x^2] &= J_x[J_z, J_x] + [J_z, J_x]J_x = i\hbar(J_xJ_y + J_yJ_x), \\ [J_z, J_y^2] &= J_y[J_z, J_y] + [J_z, J_y]J_y = -i\hbar(J_yJ_x + J_xJ_y), \\ [J_z, J_z^2] &= 0,\end{aligned}$$

using $[J_z, J_y] = -[J_y, J_z] = -i\hbar J_x$. The first two cancel, so $[J_z, \vec{J}^2] = 0$ and $[\vec{J}^2, J_z] = -[J_z, \vec{J}^2] = 0$. The same steps with x or y in place of z give the other two components. $\square$

Remark 44. $\vec{J}^2$ and J_z are compatible, so they share an eigenbasis.

A Appendix

B Solutions

Solution 1 (The exponential as a limit). *Take the logarithm of the left side, which turns the power into a product:*

$$\ln \left(1 + \frac{x}{N}\right)^N = N \ln \left(1 + \frac{x}{N}\right).$$

The Taylor series of the logarithm about $u = 0$ is

$$\ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots,$$

so with $u = x/N$, which is small once N is large,

$$N \ln \left(1 + \frac{x}{N}\right) = N \left(\frac{x}{N} - \frac{x^2}{2N^2} + \frac{x^3}{3N^3} - \cdots \right) = x - \frac{x^2}{2N} + \frac{x^3}{3N^2} - \cdots.$$

Every term after the first carries a factor $1/N$ or smaller, so they all vanish as $N \rightarrow \infty$ and the logarithm tends to x . Exponentiating,

$$\lim_{N \rightarrow \infty} \left(1 + \frac{x}{N}\right)^N = e^x.$$

Solution 2 (Other bases). *The same state has different columns in different bases:*

$$\begin{aligned}|+\hat{z}\rangle &\xrightarrow{S_z} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \\ |+\hat{z}\rangle &\xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x} | +\hat{z} \rangle \\ \langle -\hat{x} | +\hat{z} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \\ |+\hat{z}\rangle &\xrightarrow{S_y} \begin{pmatrix} \langle +\hat{y} | +\hat{z} \rangle \\ \langle -\hat{y} | +\hat{z} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}.\end{aligned}$$

Solution 3 (A probability by matrix mechanics). Write $|\psi\rangle$ as a column and $\langle +\hat{x}|$ as the conjugate row:

$$\psi \xrightarrow{S_z} \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2i \end{pmatrix}, \quad \langle +\hat{x}| \xrightarrow{S_z} \frac{1}{\sqrt{2}} (1 \quad 1).$$

Then

$$\langle +\hat{x}|\psi\rangle = \frac{1}{\sqrt{10}} (1 \quad 1) \begin{pmatrix} 1 \\ 2i \end{pmatrix} = \frac{1+2i}{\sqrt{10}}, \quad P(S_x = +\frac{\hbar}{2}) = \frac{|1+2i|^2}{10} = \frac{5}{10} = \frac{1}{2}.$$

Take the modulus of the whole sum, not of each term separately.

Solution 4 (The matrix of $\hat{S}_z$). Here every overlap is 0 or 1 by orthonormality, so only the diagonal survives:

$$\langle +\hat{z}|\hat{S}_z|+\hat{z}\rangle = \frac{\hbar}{2} (1-0) = \frac{\hbar}{2}, \quad \langle -\hat{z}|\hat{S}_z|-\hat{z}\rangle = \frac{\hbar}{2} (0-1) = -\frac{\hbar}{2},$$

with both off-diagonal entries zero, giving

$$\hat{S}_z \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

An operator is diagonal in its own basis, and the diagonal entries are its outcomes.

Solution 5 (The matrix of $\hat{S}_x$). From $|\pm\hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}}(1, \pm 1)^T$ the overlaps are $\langle +\hat{z}|\pm\hat{x}\rangle = \langle -\hat{z}|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}$ and $\langle -\hat{z}|-\hat{x}\rangle = -\frac{1}{\sqrt{2}}$, all real. Then

$$\begin{aligned} \langle +\hat{z}|\hat{S}_x|+\hat{z}\rangle &= \frac{\hbar}{2} \left(\frac{1}{2} - \frac{1}{2}\right) = 0, & \langle +\hat{z}|\hat{S}_x|-\hat{z}\rangle &= \frac{\hbar}{2} \left(\frac{1}{2} + \frac{1}{2}\right) = \frac{\hbar}{2}, \\ \langle -\hat{z}|\hat{S}_x|+\hat{z}\rangle &= \frac{\hbar}{2} \left(\frac{1}{2} + \frac{1}{2}\right) = \frac{\hbar}{2}, & \langle -\hat{z}|\hat{S}_x|-\hat{z}\rangle &= \frac{\hbar}{2} \left(\frac{1}{2} - \frac{1}{2}\right) = 0, \end{aligned}$$

where the + signs come from the single minus sign carried by $\langle -\hat{z}|-\hat{x}\rangle$ meeting the minus in front of the second projector. So

$$\hat{S}_x \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Solution 6 (The matrix of $\hat{S}_y$). Now the overlaps are complex, and the bra side is conjugated: $\langle -\hat{z}|\pm\hat{y}\rangle = \pm\frac{i}{\sqrt{2}}$ while $\langle \pm\hat{y}|-\hat{z}\rangle = \mp\frac{i}{\sqrt{2}}$, with $\langle +\hat{z}|\pm\hat{y}\rangle = \langle \pm\hat{y}|+\hat{z}\rangle = \frac{1}{\sqrt{2}}$. Then

$$\begin{aligned} \langle +\hat{z}|\hat{S}_y|-\hat{z}\rangle &= \frac{\hbar}{2} \left(\frac{1}{\sqrt{2}} \cdot \frac{-i}{\sqrt{2}} - \frac{1}{\sqrt{2}} \cdot \frac{i}{\sqrt{2}} \right) = -\frac{i\hbar}{2}, \\ \langle -\hat{z}|\hat{S}_y|+\hat{z}\rangle &= \frac{\hbar}{2} \left(\frac{i}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - \frac{-i}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \right) = +\frac{i\hbar}{2}, \end{aligned}$$

and the diagonal entries vanish because $i \cdot (-i) = 1$ in both terms. So

$$\hat{S}_y \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Solution 7 (Expectation value and uncertainty by matrix mechanics). *First move to the S_z basis:*

$$\psi \xrightarrow{S_z} \frac{1}{\sqrt{10}} \left[\begin{pmatrix} 1 \\ i \end{pmatrix} + \begin{pmatrix} 2i \\ 2 \end{pmatrix} \right] = \frac{1}{\sqrt{10}} \begin{pmatrix} 1+2i \\ 2+i \end{pmatrix}.$$

Then $\langle S_z \rangle = \psi^\dagger \hat{S}_z \psi$:

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{10} \begin{pmatrix} 1-2i & 2-i \end{pmatrix} \begin{pmatrix} 1+2i \\ -(2+i) \end{pmatrix} = \frac{\hbar}{2} \cdot \frac{5-5}{10} = 0.$$

Since $\hat{S}_z^2 = \frac{\hbar^2}{4} \mathbb{1}$, we get $\langle S_z^2 \rangle = \frac{\hbar^2}{4} \psi^\dagger \psi = \frac{\hbar^2}{4}$, and therefore

$$(\Delta S_z)^2 = \frac{\hbar^2}{4} - 0, \quad \Delta S_z = \frac{\hbar}{2}.$$

Both S_z outcomes have probability $\frac{1}{2}$ here, so the spread is maximal.

Solution 8 ($R(\phi\hat{z})$ in the S_x basis). *Apply R to the S_x basis states by expanding them in $\hat{z}$:*

$$R|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} \left(e^{-i\phi/2} |+\hat{z}\rangle \pm e^{i\phi/2} |-\hat{z}\rangle \right).$$

Hitting these with $\langle \pm\hat{x} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \pm \langle -\hat{z} |)$ gives

$$\begin{aligned} \langle +\hat{x} | R | +\hat{x} \rangle &= \langle -\hat{x} | R | -\hat{x} \rangle = \frac{1}{2} (e^{-i\phi/2} + e^{i\phi/2}) = \cos \frac{\phi}{2}, \\ \langle +\hat{x} | R | -\hat{x} \rangle &= \langle -\hat{x} | R | +\hat{x} \rangle = \frac{1}{2} (e^{-i\phi/2} - e^{i\phi/2}) = -i \sin \frac{\phi}{2}, \end{aligned}$$

using the two identities from the Taylor section. So

$$R(\phi\hat{z}) \xrightarrow{S_x} \begin{pmatrix} \cos \frac{\phi}{2} & -i \sin \frac{\phi}{2} \\ -i \sin \frac{\phi}{2} & \cos \frac{\phi}{2} \end{pmatrix}.$$

Check: at $\phi = \pi/2$, $R|+\hat{x}\rangle \xrightarrow{S_x} \frac{1}{\sqrt{2}}(1, -i)^T$, which is $|+\hat{y}\rangle$ up to a global phase, as rotating $\hat{x}$ by 90° about $\hat{z}$ should give.

Solution 9 (Expectation value from the S_x basis). *The state in the S_x basis is*

$$|\psi\rangle \xrightarrow{S_x} \psi^x = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i\sqrt{2} \end{pmatrix}.$$

Change to the S_z basis:

$$\begin{aligned} \psi^z &= R\left(\frac{\pi}{2}\hat{y}\right) \psi^x = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i\sqrt{2} \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1+i\sqrt{2} \\ 1-i\sqrt{2} \end{pmatrix}, \\ \psi^{z\dagger} &= \frac{1}{\sqrt{6}} \begin{pmatrix} 1-i\sqrt{2} & 1+i\sqrt{2} \end{pmatrix}. \end{aligned}$$

Then

$$\begin{aligned}\langle S_z \rangle &= \psi^{z\dagger} \hat{S}_z \psi^z = \frac{\hbar}{2} \cdot \frac{1}{6} \begin{pmatrix} 1 - i\sqrt{2} & 1 + i\sqrt{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 + i\sqrt{2} \\ 1 - i\sqrt{2} \end{pmatrix} \\ &= \frac{\hbar}{2} \left(\frac{3}{6} - \frac{3}{6} \right) = 0.\end{aligned}$$

Both S_z outcomes have probability $\frac{1}{2}$, since the two components have equal modulus.

Solution 10 ($\hat{S}_z$ in the S_x basis). Here the eigenbasis of $\hat{S}_z$ is $\{|\pm\hat{z}\rangle\}$ and the target basis is $\{|\pm\hat{x}\rangle\}$, so $U_{j\alpha} = \langle \pm\hat{z} | \pm\hat{x} \rangle$, which is $R\left(\frac{\pi}{2}\hat{y}\right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. It is real and symmetric, so $U^\dagger = U$, and

$$\begin{aligned}\hat{S}_z &\xrightarrow{S_x} U^\dagger D U = \frac{\hbar}{2} \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \frac{\hbar}{4} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.\end{aligned}$$

This is $\frac{\hbar}{2}\sigma_x$: in the S_x basis, $\hat{S}_z$ looks the way $\hat{S}_x$ looks in the S_z basis.

Now keep the state in the S_x basis, $\psi^x = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i\sqrt{2} \end{pmatrix}$:

$$\begin{aligned}\langle S_z \rangle &= \psi^{x\dagger} \hat{S}_z \psi^x = \frac{\hbar}{2} \cdot \frac{1}{3} \begin{pmatrix} 1 & -i\sqrt{2} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i\sqrt{2} \end{pmatrix} \\ &= \frac{\hbar}{2} \cdot \frac{1}{3} \begin{pmatrix} 1 & -i\sqrt{2} \end{pmatrix} \begin{pmatrix} i\sqrt{2} \\ 1 \end{pmatrix} = \frac{\hbar}{2} \cdot \frac{i\sqrt{2} - i\sqrt{2}}{3} = 0,\end{aligned}$$

matching the answer found by changing the state's basis. In general, for $\psi^x = (c_+, c_-)^T$ this gives $\langle S_z \rangle = \hbar \operatorname{Re}(c_+^* c_-)$, which vanishes here because the relative phase between the components is i .

C Useful Links

Stern-Gerlach experiment:

<https://www.youtube.com/shorts/QG7kJXxN1Kk>